

Polyphenols and phenolic acids from strawberry and apple decrease glucose uptake and transport by human intestinal Caco-2 cells.

The effect of polyphenols, phenolic acids and tannins (PPTs) from strawberry and apple on uptake and apical to basolateral transport of glucose was investigated using Caco-2 intestinal cell monolayers. Substantial inhibition on both uptake and transport was observed by extracts from both strawberry and apple. Using sodium-containing (glucose transporters SGLT1 and GLUT2 both active) and sodium-free (only GLUT2 active) conditions, we show that the inhibition of GLUT2 was greater than that of SGLT1. The extracts were analyzed and some of the constituent PPTs were also tested. Quercetin-3-O-rhamnoside ($IC_{50}=31\text{ }\mu\text{M}$), phloridzin ($IC_{50}=146\text{ }\mu\text{M}$), and 5-caffeoylquinic acid ($IC_{50}=2570\text{ }\mu\text{M}$) contributed 26, 52 and 12%, respectively, to the inhibitory activity of the apple extract, whereas pelargonidin-3-O-glucoside ($IC_{50}=802\text{ }\mu\text{M}$) contributed 26% to the total inhibition by the strawberry extract. For the strawberry extract, the inhibition of transport was non-competitive based on kinetic analysis, whereas the inhibition of cellular uptake was a mixed-type inhibition, with changes in both $V(\text{max})$ and apparent $K(\text{m})$. The results in this assay show that some PPTs inhibit glucose transport from the intestinal lumen into cells and also the GLUT2-facilitated exit on the basolateral side. Copyright © 2010 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.